H M PREMA

premahiremath2005@gmail.com |+91 7760051573| Bengaluru.

EDUCATION

B.E in Electronics and Communication **2023 - Present**
Sir M. Visvesvaraya Institute of Technology, Bangalore
CGPA: 8.5

SMT Yalamanchili Janata Rani PU Collage, Gangavathi **2020 - 2022**
PCMB | 92%

VidyaSagara Residential Public School ,Kampli **2020**

State Board 10th Result | 96%

CAREER OBJECTIVE

Electronics and Communication Engineering student seeking an opportunity to apply academic knowledge, gain practical exposure, and enhance technical skills in core electronics and related domains.

CERTIFICATES

- Completed 5-Day PCB Workshop at ComedKares Innovation Hub, Yelahanka Center (March 2025)-Hands-on training in PCB design and development.
- AI for Students: Build Your Own Generative AI Model – NxtWave (Oct 2024) Learned fundamentals of Generative AI and model-building techniques.
- IoT Innovators: Hands-on Workshop with Node MCU ESP32 – Sir MVIT, IEEE Photonics Society (Sept–Oct 2024) Gained practical knowledge in IoT system design and Node MCU programming.

PROJECTS

Smart Shoe for the Blind

- Developed an assistive system using Arduino UNO and ultrasonic sensors.
- Detected nearby obstacles and provided alerts using a buzzer/vibration motor.
- Built and tested a working hardware prototype.

ResiScan – AI-Powered Resistor Scanner

- Created a custom dataset of 800+ resistor images, manually captured and labeled with corresponding resistance values and 4-band color codes.
- Developed a web-based system using MobileNetv2(TensorFlow) to classify resistor color bands and predict resistance values in real time.

Oral Cancer Detection Using Histopathological Images

- Built a MobileNetV2-based CNN model to classify oral cancer from histopathological images using Kaggle dataset
- Applied preprocessing, data augmentation, and hyperparameter tuning; evaluated performance using confusion matrix and ROC curve

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SKILLS

- Programming Languages: Embedded C, Python(basics), Verilog, Arduino Programming, java(basics)
- Tools: Keil, Multisim, MATLab, LabVIEW
- Soft Skills: Communication, Problem Solving, Team Collaboration

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